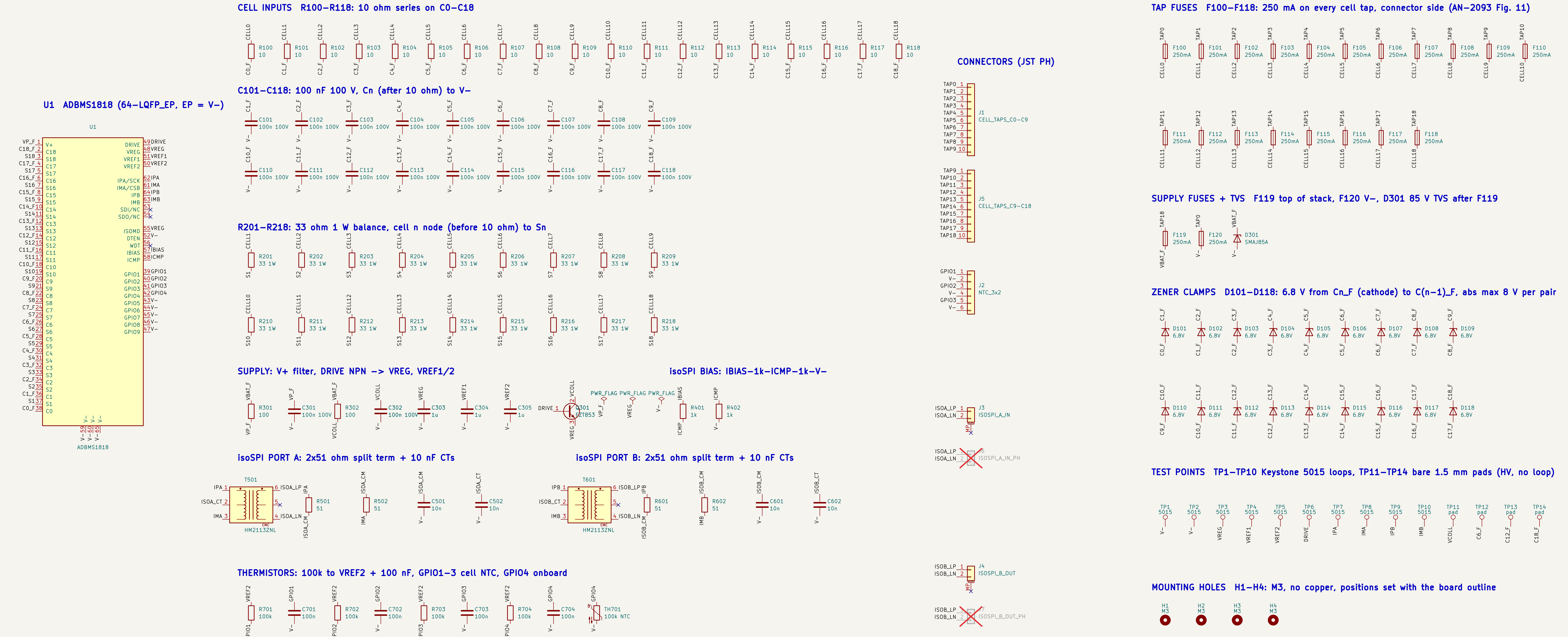
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.



CELL INPUTS R100–R118: 10 ohm series on C0–C18

C101–C118: 100 nF 100 V, Cn (after 10 ohm) to V−

U1 ADBMS1818 (64-LQFP_EP, EP = V−)

R201–R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn

SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2

isoSPI BIAS: IBIAS–1k–ICMP–1k–V−

isoSPI PORT A: 2x51 ohm split term + 10 nF CTs

isoSPI PORT B: 2x51 ohm split term + 10 nF CTs

THERMISTORS: 100k to VREF2 + 100 nF, GPIO1–3 cell NTC, GPIO4 onboard

CONNECTORS (JST PH)

TAP FUSES F100–F118: 250 mA on every cell tap, connector side (AN-2093 Fig. 11)

SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119

ZENER CLAMPS D101–D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair

TEST POINTS TP1–TP10 Keystone 5015 loops, TP11–TP14 bare 1.5 mm pads (HV, no loop)

MOUNTING HOLES H1–H4: M3, no copper, positions set with the board outline

U1
ADBMS1818
V+ C18 S18 C17 S17 C16 S16 C15 S15 C14 S14 C13 S13 C12 S12 C11 S11 C10 S10 C9 S9 C8 S8 C7 S7 C6 S6 C5 S5 C4 S4 C3 S3 C2 S2 C1 S1 C0
DRIVE VREG VREF1 VREF2 IPA/SCK IMA/CSB IPB IMB SDI/NC SDO/NC ISOMD DTEN WDT IBIAS ICMP GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9
VP_F C18_F S18 C17_F S17 C16_F S16 C15_F S15 C14_F S14 C13_F S13 C12_F S12 C11_F S11 C10_F S10 C9_F S9 C8_F S8 C7_F S7 C6_F S6 C5_F S5 C4_F S4 C3_F S3 C2_F S2 C1_F S1 C0_F
DRIVE VREG VREF1 VREF2 IPA IMA IPB IMB VREG V− IBIAS ICMP GPIO1 GPIO2 GPIO3 GPIO4 V− V− V− V− V−
R100 10 R101 10 R102 10 R103 10 R104 10 R105 10 R106 10 R107 10 R108 10 R109 10 R110 10 R111 10 R112 10 R113 10 R114 10 R115 10 R116 10 R117 10 R118 10
CELL0–CELL18
C0_F–C18_F
C101–C118 100n 100V
R201–R218 33 1W
F100–F118 250mA
TAP0–TAP18
J1 CELL_TAPS_C0–C9
J5 CELL_TAPS_C9–C18
J2 NTC_3x2
GPIO1 V− GPIO2 V− GPIO3 V−
J3 ISOSPI_A_IN
ISOA_LP ISOA_LN
ISOSPI_A_IN_PH
J4 ISOSPI_B_OUT
ISOB_LP ISOB_LN
ISOSPI_B_OUT_PH
F119 250mA
F120 250mA
D301 SMAJ85A
VBAT_F
D101–D109 6.8V
D110–D118 6.8V
R301 100
C301 100n 100V
R302 100
C302 100n 100V
C303 1u
C304 1u
C305 1u
Q301 FZT853
VBAT_F VP_F VCOLL VREG VREF1 VREF2 DRIVE
PWR_FLAG PWR_FLAG PWR_FLAG
R401 1k
R402 1k
IBIAS ICMP
T501 HM2113ZNL
IPA ISOA_CT IMA ISOA_LP ISOA_LN
R501 51
R502 51
C501 10n
C502 10n
ISOA_CM
T601 HM2113ZNL
IPB ISOB_CT IMB ISOB_LP ISOB_LN
R601 51
R602 51
C601 10n
C602 10n
ISOB_CM
R701 100k C701 100n R702 100k C702 100n R703 100k C703 100n R704 100k C704 100n
TH701 100k NTC
TP1–TP10 5015
TP11–TP14 pad
V− V− VREG VREF1 VREF2 DRIVE IPA IMA IPB IMB VCOLL C6_F C12_F C18_F
H1 M3 H2 M3 H3 M3 H4 M3

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

Size: User | Date: 2026−11−06 | **Rev: 0−draft**
KiCad E.D.A. 10.0.6 | Id: 1/1